Guten Morgen!

AZ-104

Administer Azure Storage

Tag 3

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AZ-104 Agenda

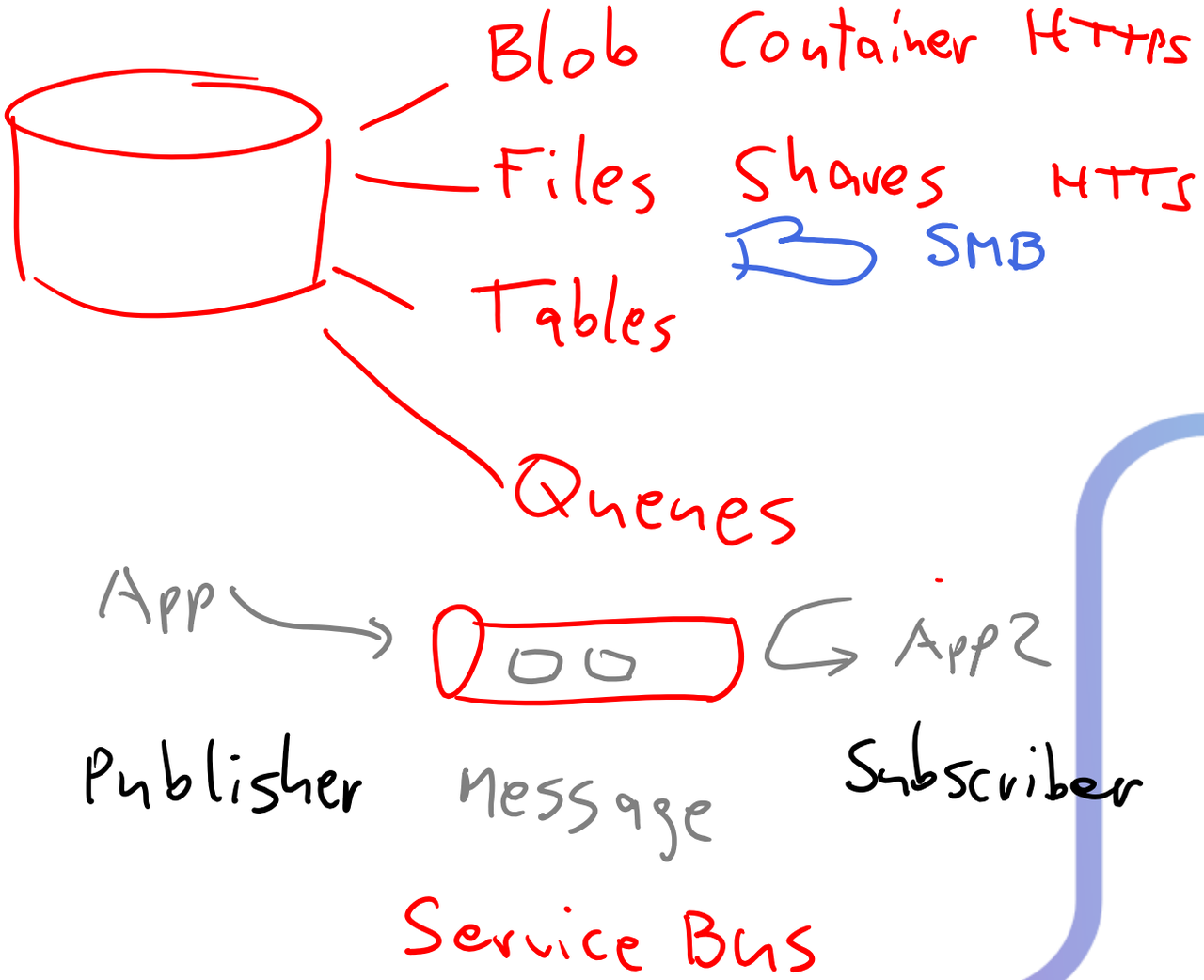
01: Administer Identity
02: Administer Governance and Compliance
03: Administer Azure Resources
04: Administer Virtual Networking
05: Administer Intersite Connectivity
06: Administer Network Traffic Management
07: Administer Azure Storage
08: Administer Azure Virtual Machines
09: Administer PaaS Compute Options
10: Administer Data Protection
11: Administer Monitoring

www App Service
Tim Berners-Lee CERN

Learning Objectives - Administer Azure Storage

- Configure Storage Accounts v2.0
- Configure Blob Storage
- Configure Storage Security
- Configure Azure Files
- Lab 07 – Manage Azure Storage

RBAC Roles
SAS
Key



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Storage Account Standard → HDD
Premium → SSD

Configure Storage Accounts

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Learning Objective - Configure Storage Accounts

- Explore Azure Storage Services
- Determine Storage Account Kinds
- Determine Replication Strategies
- Access Storage
- Secure Storage Endpoints
- Demonstration – Configure a storage account
- Learning Recap

Exam objectives

Configure and manage storage accounts

- Create and configure storage accounts
- Configure Azure Storage redundancy

Configure access to storage

- Configure Azure Storage firewalls and virtual networks

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Explore Azure Storage Services

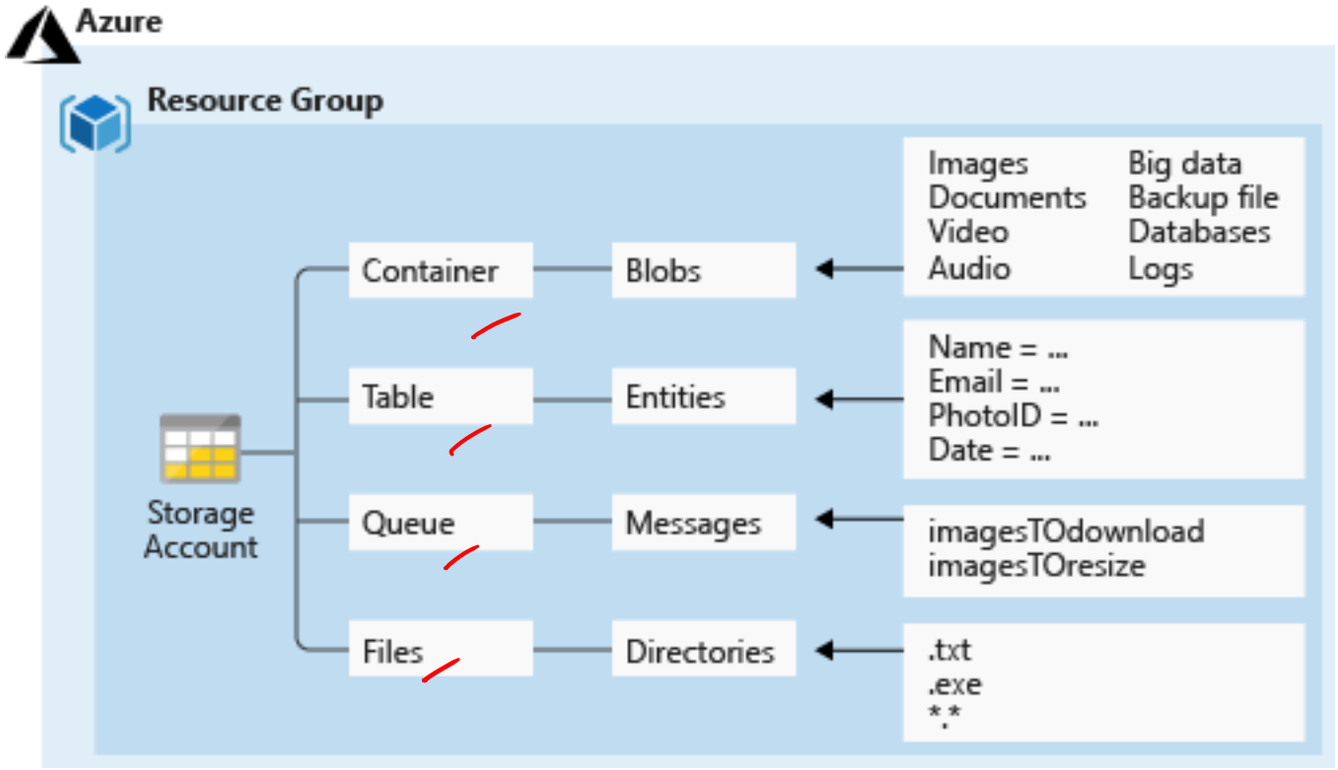
A service that you can use to store files, messages, tables, and other types of information

Azure Containers: A massively scalable object store for text and binary data

Azure Tables: Ideal for storing structured, non-relational data

Azure Queues: A messaging store for reliable messaging between application components

Azure Files: Managed file shares for cloud or on-premises deployments



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Determine Storage Account Kinds

All storage accounts are encrypted using Storage Service Encryption (SSE) for data at rest

Storage Account	Recommended usage
Standard general-purpose v2	Most scenarios including Blob, File, Queue, Table, and Data Lake Storage.
Premium block blobs	Block blob scenarios with high transactions rates, or scenarios that use smaller objects or require consistently low storage latency.
Premium file shares	Enterprise or high-performance file share applications.
Premium page blobs	Premium high-performance page blob scenarios.

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Determine Replication Strategies

	Primary Region	Secondary Region
A node within a data center becomes unavailable	Locally redundant storage	Geo- redundant storage
An entire data center becomes unavailable	Zone redundant storage	Geo-zone redundant storage

A region-wide outage occurs in the primary region

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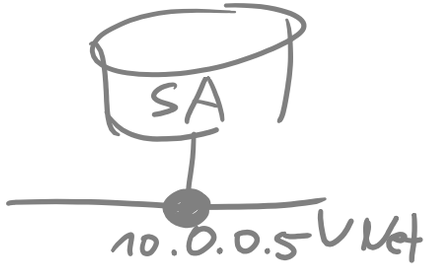
Access Storage

Every object has a unique URL address – based on account name and storage type

Container service: https://*mystorageaccount*.blob.core.windows.net
Table service: https://*mystorageaccount*.table.core.windows.net
Queue service: https://*mystorageaccount*.queue.core.windows.net
File service: https://*mystorageaccount*.file.core.windows.net

IANA → Public IP
20.1.2.3



private DNS Zone → Private IP
Private EP

- If you prefer you can configure a custom domain name

CNAME record	Target
blobs.contoso.com	contosoblobs.blob.core.windows.net

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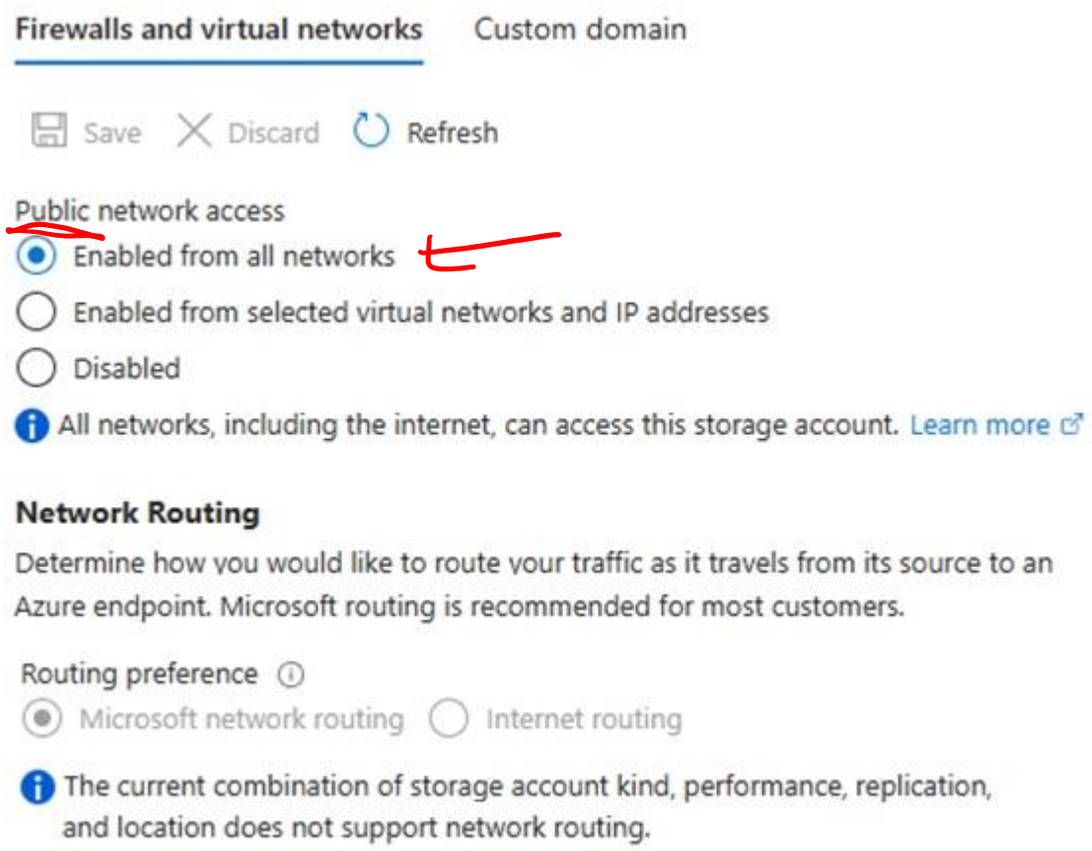
Secure Storage Endpoints

Firewalls and Virtual Networks restrict access to the Storage Account from specific Subnets on Virtual Networks or public IP's

Subnets and Virtual Networks must exist in the same Azure Region or Region Pair as the Storage Account



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Configure Blob Storage

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Learning Objective - Blob Storage

- Implement Blob Storage
- Create Blob Containers
- Create Blob Access Tiers
- Add Blob Lifecycle Management Rules
- Determine Blob Object Replication
- Demonstration – Configure Blob Storage
- Learning Recap

Exam objectives

Configure Azure Files and Azure Blob Storage

- Create and configure a container in Blob storage
- Configure storage tiers
- Configure blob lifecycle management
- Configure blob versioning

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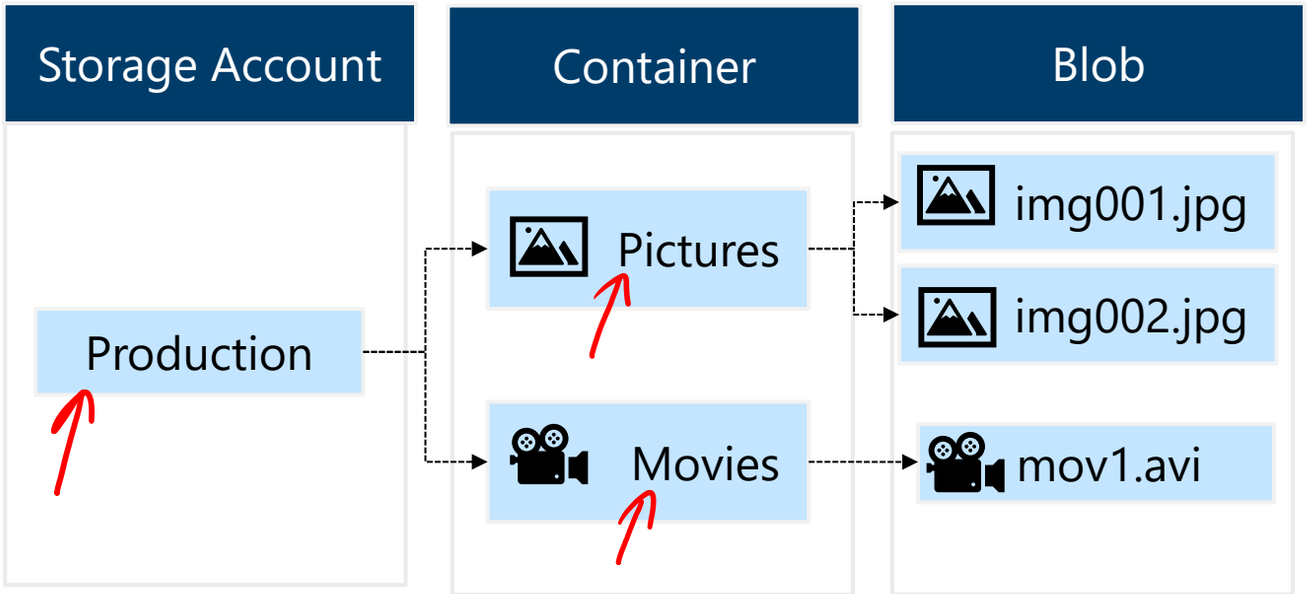
Implement Blob Storage

- Stores unstructured data in the cloud
- Can store any type of text or binary data
- Also referred to as *object storage*

Common uses:

- Serving images or documents directly to a browser
- Storing files for distributed access
- Streaming video and audio
- Storing data for backup and restore, disaster recovery, archiving
- Storing data for analysis by an on-premises or Azure-hosted service



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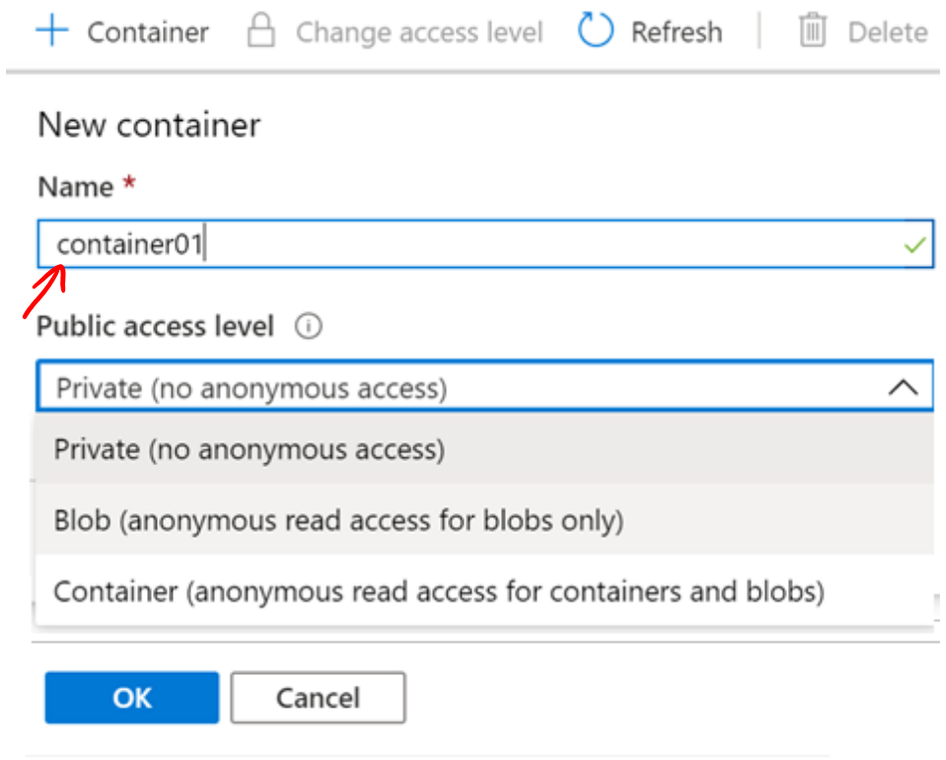
Create Blob Containers

- All blobs must be in a container
- Accounts have unlimited containers
- Containers can have unlimited blobs
- Restrict access using the public access level



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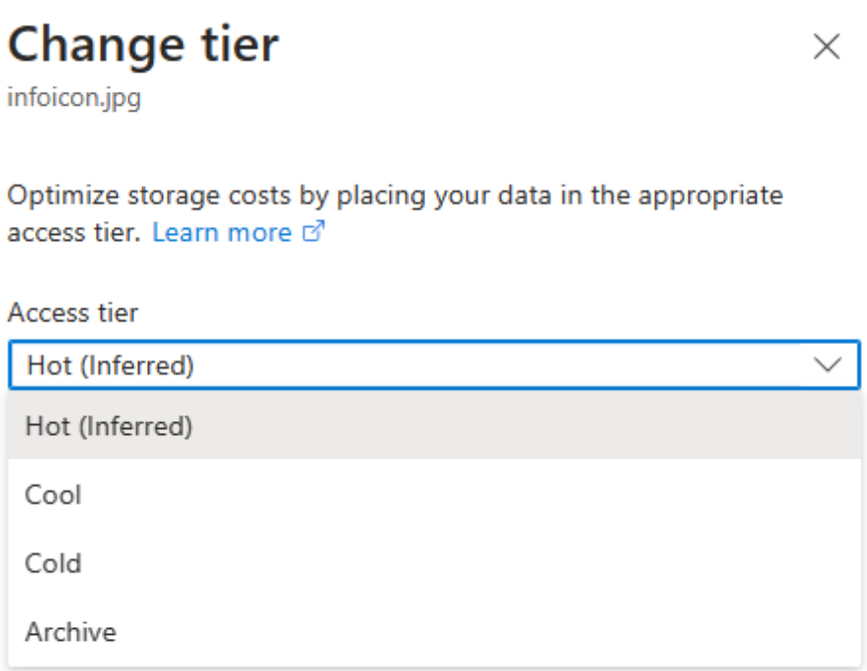
Create Blob Access Tiers

Hot tier – Data that is accessed or modified frequently

Cool tier – Data that is infrequently accessed or modified and stored for at least 30 days

Cold tier – Data that is infrequently accessed or modified and stored for at least 90 days

Archive – Data that can tolerate several hours of retrieval latency and will remain in the Archive tier for at least 180 days



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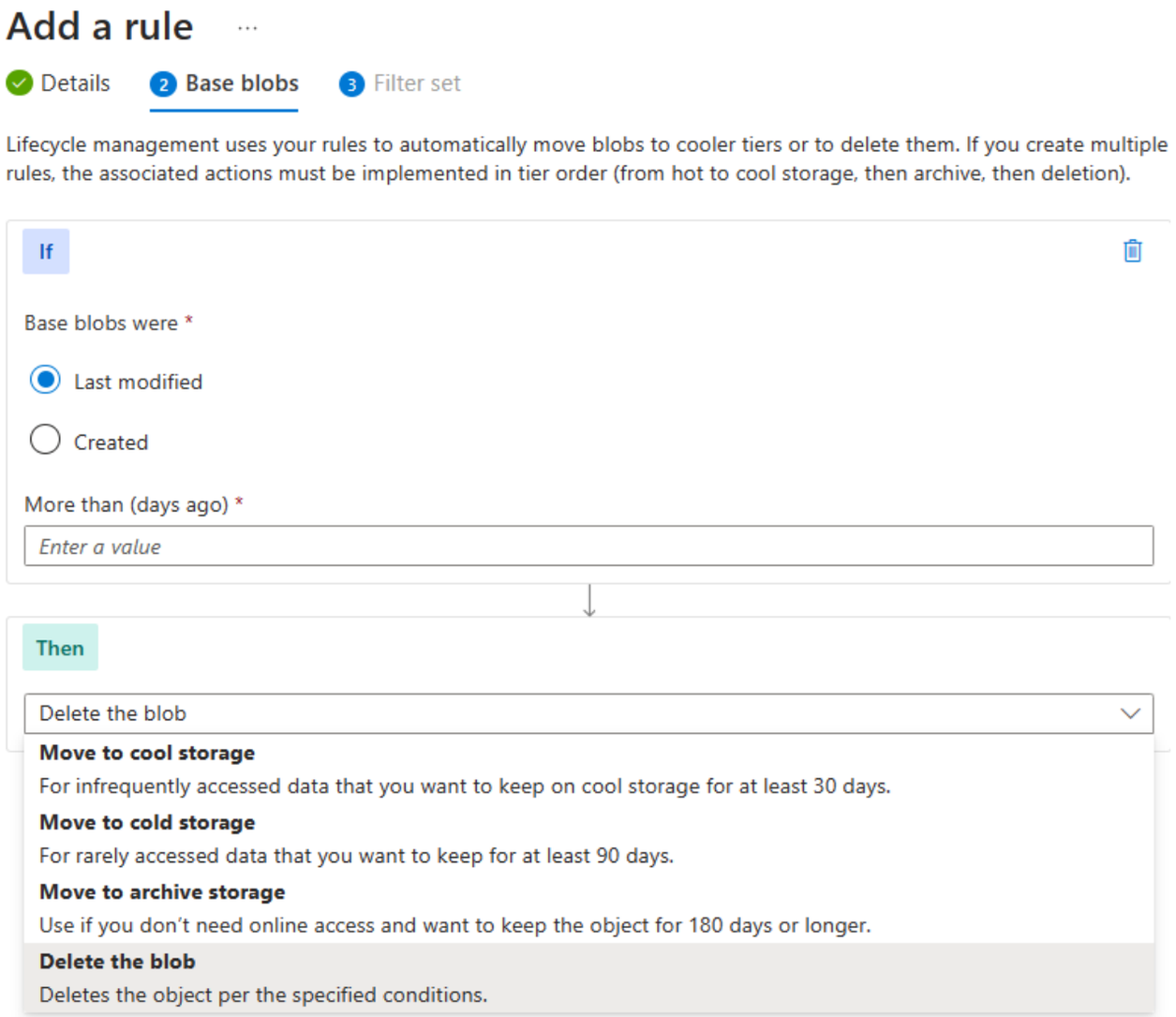
Add Blob Lifecycle Management Rules

- Transition blobs to a cooler storage tier to optimize for performance and cost
- Delete blobs at the end of their lifecycle
- Apply rules for lifecycle management tasks



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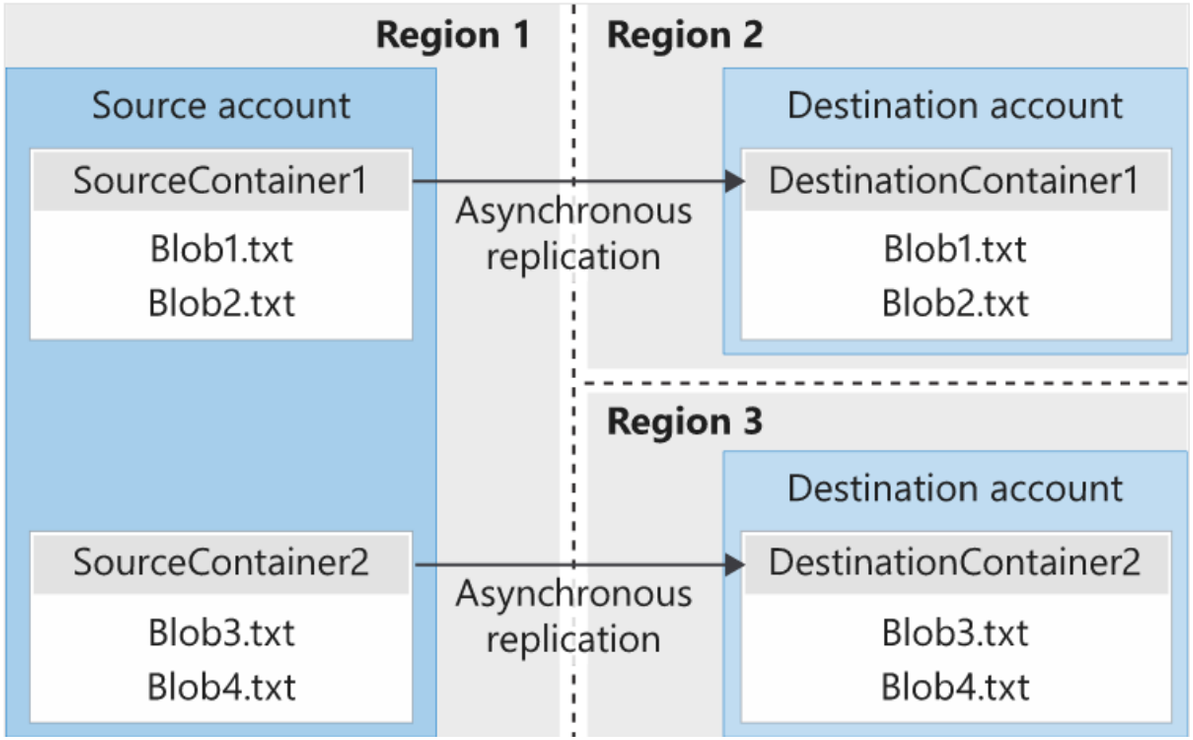
Determine Blob Object Replication

- Asynchronous to any other Region
- Minimizes latency for read requests
- Increases efficiency for compute workloads
- Optimizes data distribution
- Optimizes costs



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Configure Storage Security

Container

Anonymous Access No

Private Access Yes

- key
- SAS
- RBAC

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Learning Objective - Configure Storage Security

- Review Storage Security Strategies
- Create Shared Access Signatures
- Identify URI and SAS Parameters
- Demonstration – Configure Storage Security
- Determine Storage Service Encryption
- Create Customer Managed Keys
- Appy Storage Security Best Practices
- Learning Recap

Exam objectives

Configure access to storage

- Create and use shared access signature (SAS) tokens
- Configure stored access policies
- Manage access keys

Configure and manage storage accounts

- Configure storage account encryption

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Review Storage Security Strategies

Storage Service Encryption

Authentication with Entra ID and RBAC

Client-side encryption, HTTPS, and SMB 3.0 for data in transit

Azure disk encryption

Shared Access Signatures – delegated access SAS

Shared Key – encrypted signature string

Anonymous access to containers and blobs

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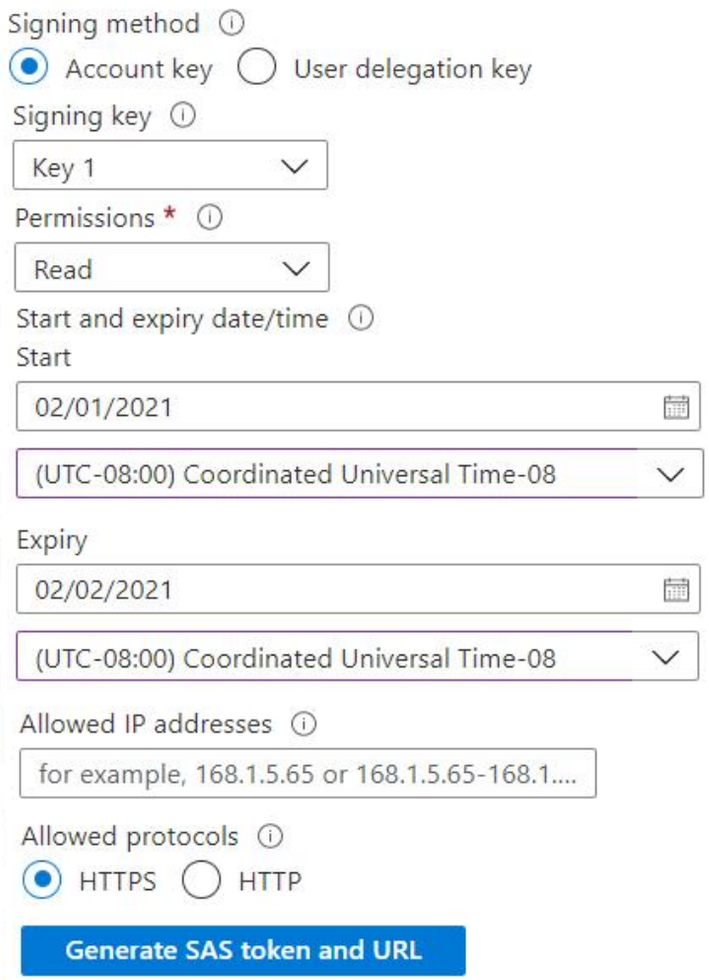
Create Shared Access Signatures

- Provides delegated access to resources
- Grants access to clients without sharing your storage account keys
- The account SAS delegates access to resources in one or more of the storage services
- The service SAS delegates access to a resource in just one of the storage services



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Identify URI and SAS Parameters

- A SAS is a signed URI that points to one or more storage resources
- Consists of a storage resource URI and the SAS token

```
https://myaccount.blob.core.windows.net/?sp=r&st=2020-05-
11T18:31:43Z&se=2020-05-12T02:31:43Z&spr=https&sv=2019-10-
10&sr=b&sig=jOqABJZHfUVeBQ3yVn7kWiCKlO0sxCiK1rzEchfAz8U%3D
```

Includes parameters for resource URI, storage services version, services, resource types, start time, expiry time, resource, permissions, IP range, protocol, signature

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Demonstration – Configure storage security

- Generate a shared access signature
- Review the configuration permissions
- Use the SAS URL to test the permissions

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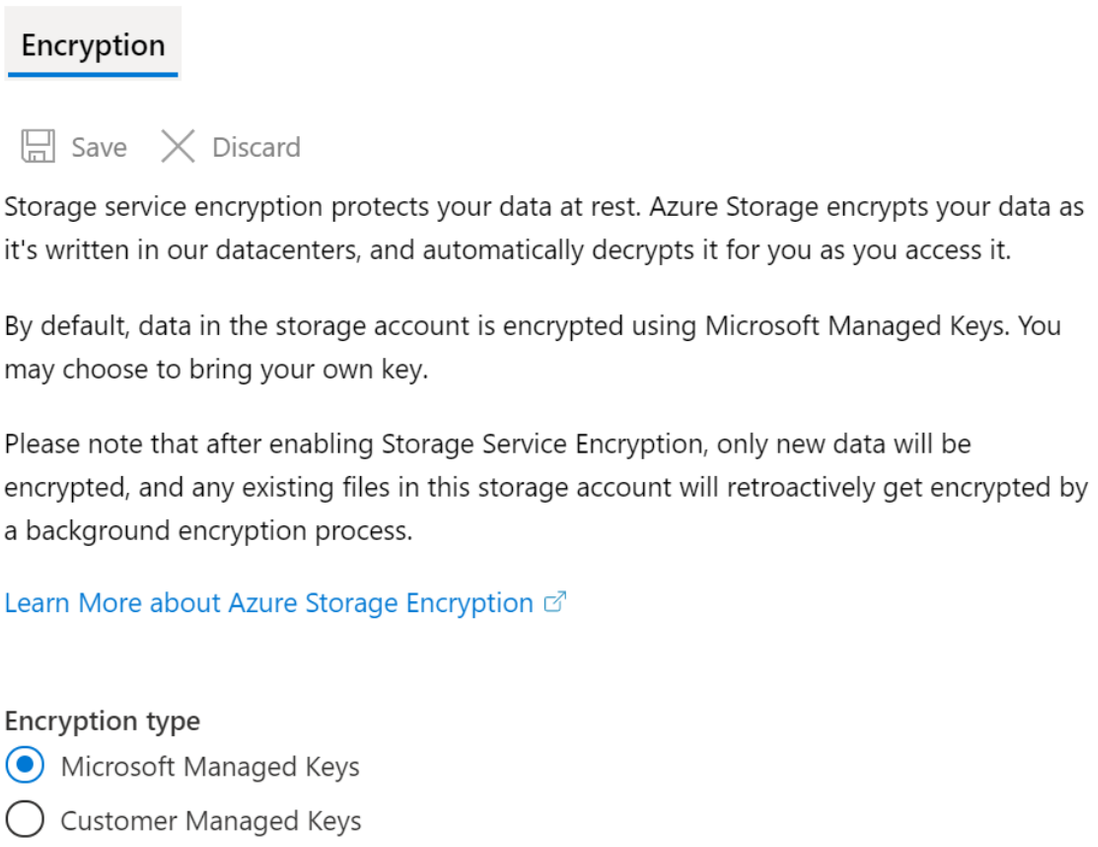
Determine Storage Service Encryption

You can use your own key (next topic)

- Protects your data for security and compliance
- Automatically encrypts and decrypts your data
- Encrypted through 256-bit AES encryption
- Is enabled for all new and existing storage accounts and cannot be disabled
- Is transparent to users



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Create Customer Managed Keys

RSA
EC

- Use the Azure Key Vault to manage your encryption keys
- Create your own encryption keys and store them in a key vault
- Use Azure Key Vault's APIs to generate encryption keys
- Custom keys give you more flexibility and control

Encryption type

- ○ Microsoft Managed Keys
- ◉ Customer Managed Keys

The storage account named 'storage987123' will be granted access to the selected key vault. Both soft delete and purge protection will be enabled on the key vault and cannot be disabled. Learn more about customer managed keys

Encryption key

- ○ Enter key URI
- ◉ Select from Key vault

Key vault and key *

Key vault: keyvault987123
Key: storagekey
Select a key vault and key

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Apply Storage Security Best Practices

- Always use HTTPS to create or distribute a SAS
- Reference stored access policies where possible
- Use near-term expiration times on an ad hoc SAS
- Use Storage Analytics to monitor your application
- Be specific with the resource to be accessed
- Understand that your account will be billed for any usage
- Validate data written using SAS
- Don't assume SAS is always the correct choice

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Configure Azure Files

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"File Server" ~~DAT~~

Learning Objective - Configure Azure Files

Share perm. = RBAC Role

File
ACL
NTFS Perm

- Compare Storage for File Shares and Blob Data
- Manage File Shares
- Create File Share Snapshots
- Implement Soft Delete for Azure Files
- Use Azure Storage Explorer
- Demonstration – Configure File Shares
- Learning Recap

Exam objectives

Configure Azure Files and Azure Blob Storage

- Create and configure a file share in Azure Storage
- Configure snapshots and soft delete for Azure Files
- Manage data by using Azure Storage Explorer

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Compare Storage for File Shares and Blob Data

Feature	Description	When to use
Azure Files	SMB interface, client libraries, and a REST interface that allows access from anywhere to stored files	• Lift and shift an application to the cloud • Store shared data across multiple virtual machines • Store development and debugging tools that need to be accessed from many virtual machines
Azure Blobs	Client libraries and a REST interface that allows unstructured data (flat namespace) to be stored and accessed at a massive scale in block blobs	• Support streaming and random-access scenarios • Access application data from anywhere

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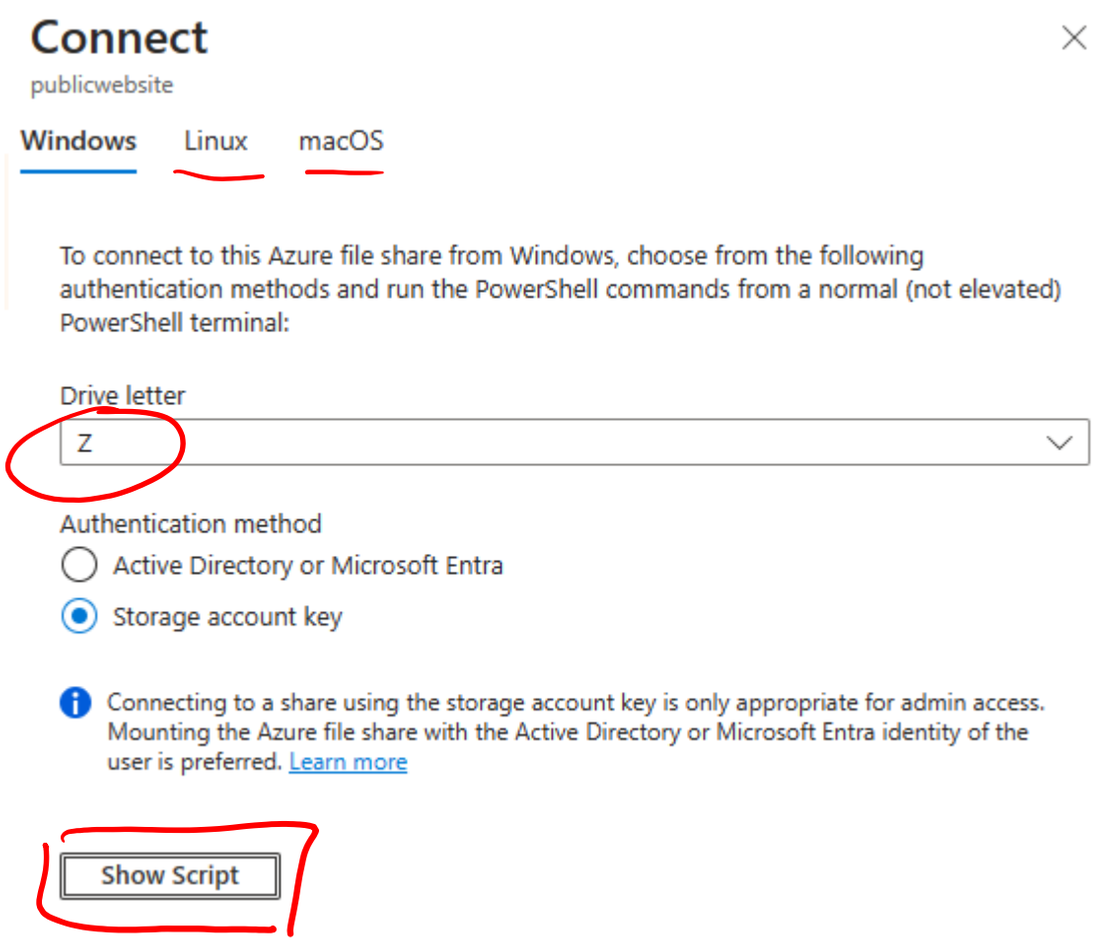
Manage File Shares

- Windows – ensure port 445 is open
- Linux – mount the drive
- MacOS – mount the drive
- Secure transfer required – SMB 3.0 encryption

3.1



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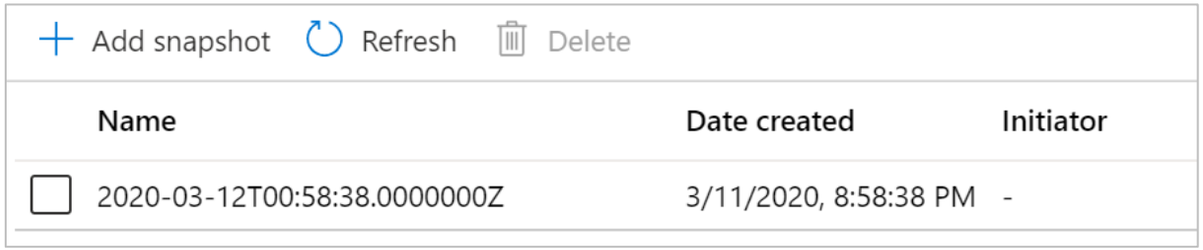
Create File Share Snapshots

- Protection against application error and data corruption
- Protection against accidental deletions or unintended changes
- Support backup and recovery

- Incremental snapshot that captures the share state at a point in time
- Snapshot at the *file share level*, and restore at the *file level*
- Is read-only copy of your data



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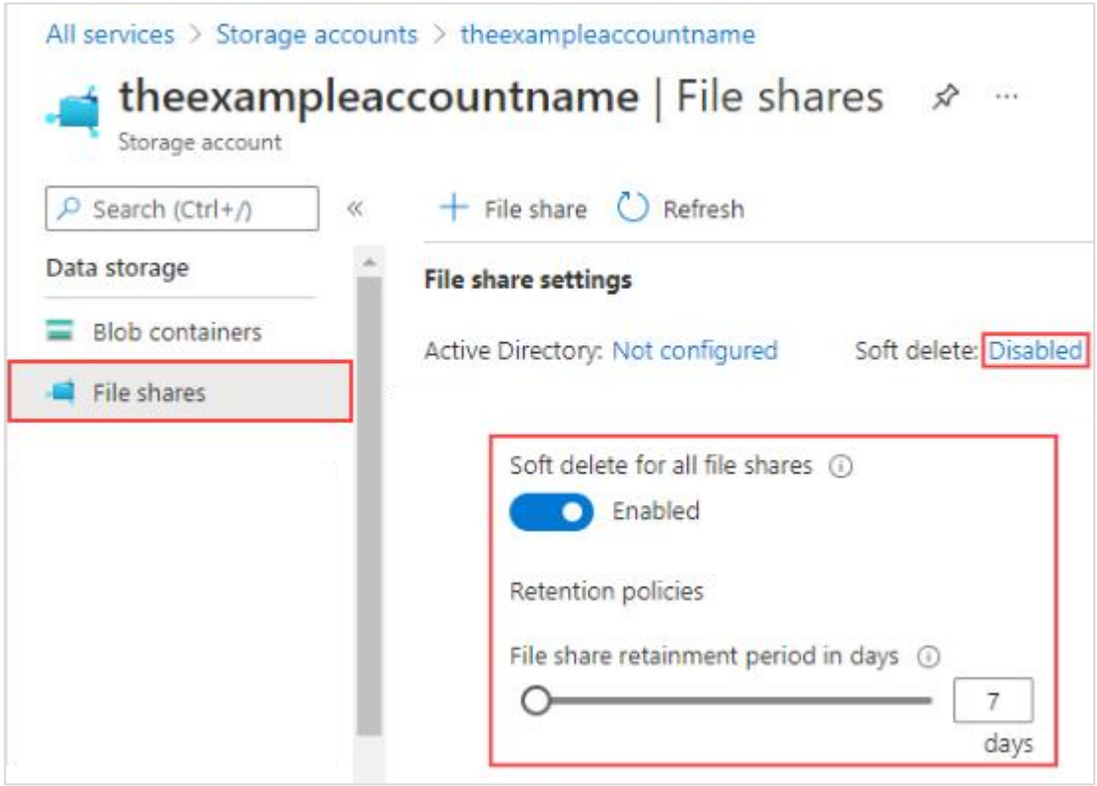
Implement Soft Delete for Azure Files

- Recovery from accidental data loss
- Major change or upgrade scenarios
- Business continuity – ransomware situations
- Data compliance retention

- Enabled at the storage account level
- Transitions content to a soft deleted state
- Provides a retention period of 1 and 365 days
- Works on new or existing file shares
- Doesn't work for NFS shares



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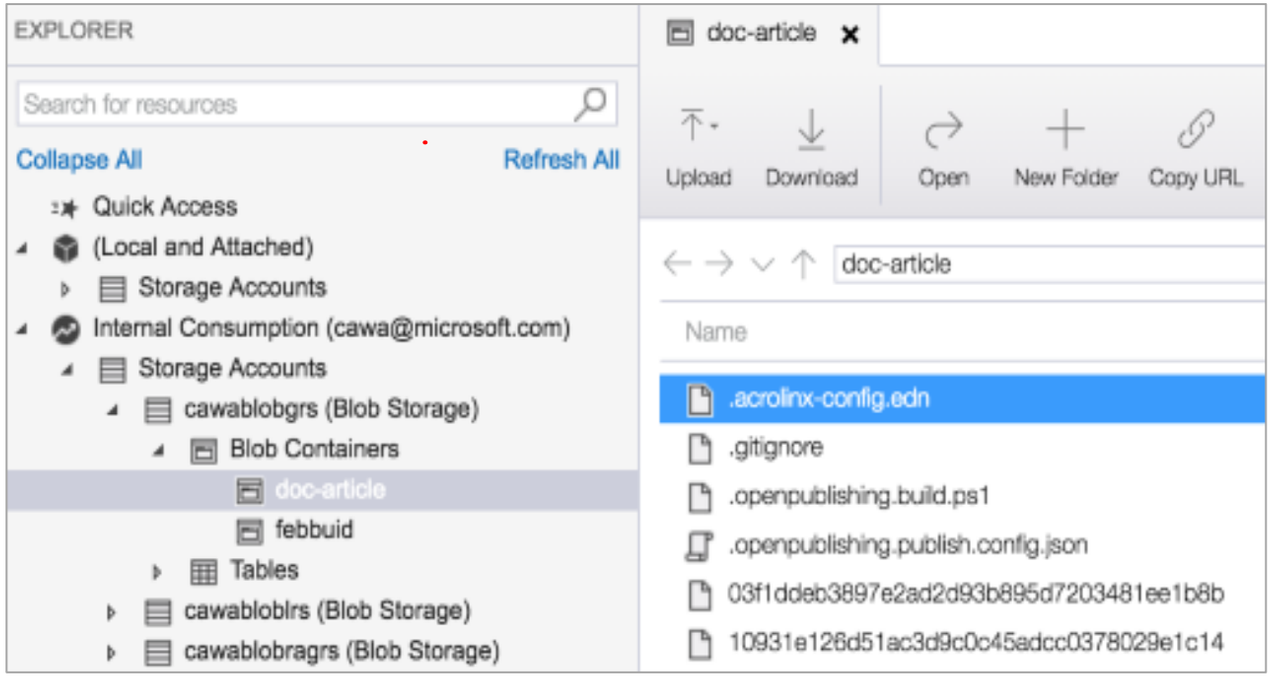
Tool

Win
Linux
MacOS

Use Azure Storage Explorer

azcopy

- Download and install
- Access multiple accounts and subscriptions
- Create, delete, view, edit storage resources
- View and edit Blob, Queue, Table, File, Cosmos DB storage and Data Lake Storage
- Obtain shared access signature (SAS) keys
- Available for Windows, Mac, and Linux



Also consider portal-based Azure Storage Browser and Azure Storage Mover

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Lab – Manage Azure Storage

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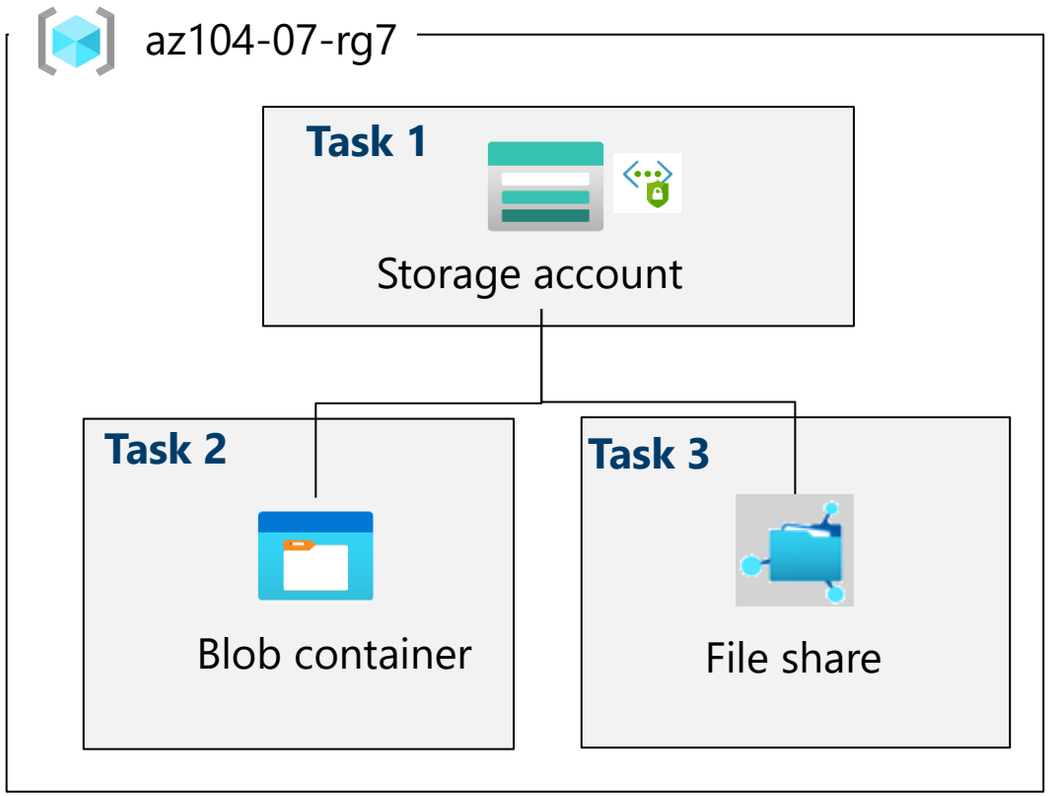
Lab 07 – Architecture diagram



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